

MOBILE MANUFACTURER DATA ANALYSIS

SQL ANALYSIS

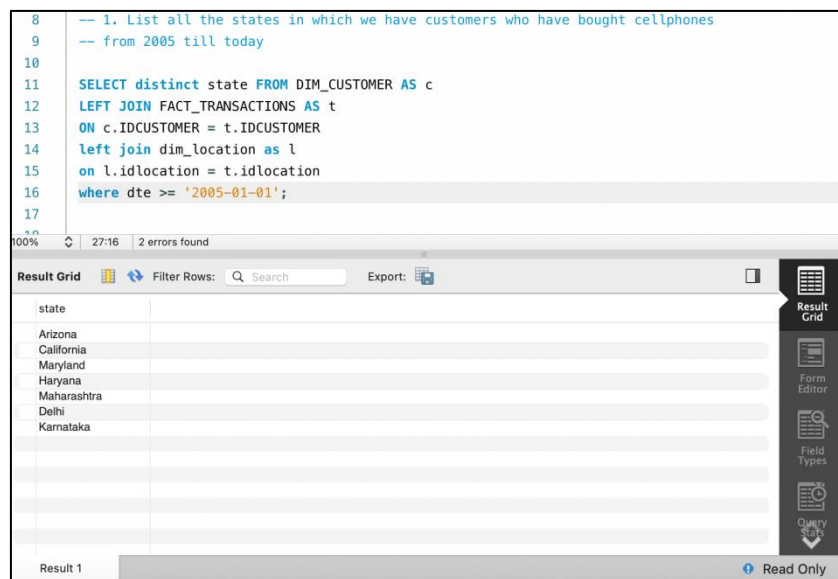
Q1 Which states have customers who purchased cellphones from 2005 onwards?

SQL QUERY

```
Select distinct state from dim_customer as c
Left join fact_transactions as t
On c.idcustomer = t.idcustomer
left join dim_location as l
on l.idlocation = t.idlocation
where dte >= '2005-01-01'
```

RESULT

Arizona, Maryland, Haryana, Maharashtra, Delhi and Karnataka are the states with customers who purchased cellphones from 2005 onwards.



ANALYSIS

The analysis shows that cellphone purchases were recorded across six states, giving an overview of the geographical spread of the customer base.

Q2 Which US state has the highest number of Samsung cellphone purchases?

SQL QUERY

```

select state from fact_transactions as t
left join dim_location as l
on l.idlocation = t.idlocation
left join dim_model as M
on T.idmodel= M.idmodel
left join dim_manufacturer as A
on M.idmanufacturer = A.idmanufacturer
where country = 'US'
and manufacturer_name= 'SAMSUNG'
group by state, manufacturer_name
order by sum(quantity) desc
limit 1

```

RESULT

Arizona is the US state with highest number of Samsung cellphone purchases.

```

25 -- What state in the US is buying the most 'Samsung' cell phones?
26
27 • select state FROM FACT_TRANSACTIONS AS t
28 left join dim_location as l
29 on l.idlocation = t.idlocation
30 left join DIM_MODEL AS M
31 on T.IDMODEL = M.IDMODEL
32 left join DIM_MANUFACTURER AS A
33 on M.IDMANUFACTURER= A.IDMANUFACTURER
34 where l.COUNTRY = 'US'
35 and MANUFACTURER_NAME = 'SAMSUNG'
36 group by state, manufacturer_name
37 order by SUM(QUANTITY) DESC
38 limit 1;
39
40
41 -- Q2--END
42

```

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Result Grid Filter Rows: Search Export: Fetch rows:

state
Arizona

Result 4

ANALYSIS

The analysis shows that Arizona has the highest number of Samsung cellphone purchases among US states, indicating strong demand for Samsung products in this market.

Q3 How many transactions were made for each model by zip code and state?

SQL QUERY

```

Select idmodel, zipcode, state, count(*)
As tran_cnt from fact_transactions as t
left join dim_location as l
on l.idlocation = t.idlocation
Group by idmodel, zipcode, state
Order by idmodel, state

```

RESULT

The result gives the Idmodel, zipcode, state and transaction count. The result have total 188 rows.

```
45 -- Show the number of transactions for each model per zip code per state.
46
47 • SELECT IDMODEL, ZIPCODE, STATE, COUNT(*) AS TRAN_CNT FROM FACT_TRANSACTIONS AS t
48   LEFT JOIN dim_location as l
49   ON l.idlocation = t.idlocation
50   GROUP BY IDMODEL, ZIPCODE, STATE
51   ORDER BY IDMODEL, state;
52
```

IDMODEL	ZIPCODE	STATE	TRAN_CNT
101	85000	Arizona	2
101	85117	Arizona	1
101	92703	California	3
101	94005	California	1
101	110004	Delhi	4
101	122002	Haryana	3
101	530068	Karnataka	1
101	400006	Maharashtra	1
101	21163	Maryland	1
101	21648	Maryland	4
102	85117	Arizona	1
102	92703	California	1
102	110004	Delhi	1
102	122002	Haryana	1
102	530068	Karnataka	3
103	94005	California	1
103	110004	Delhi	2
103	122002	Haryana	1
103	21648	Maryland	3

ANALYSIS

The analysis helps identify the locations where different cellphone models are being purchased, allowing the business to understand the geographic distribution of its sales.

Q4 Which is the cheapest cellphone and what is its price?

SQL QUERY

```
select distinct model_name, unit_price from fact_transactions as t
left join dim_model as m
on t.idmodel = m.idmodel
order by unit_price
limit 1;
```

RESULT

The cheapest model_name is 3210 and it's unit price is 14.00.

```

52
53
54 -- Q4--BEGIN
55
56 -- 4. Show the cheapest cellphone (Output should contain the price also)
57
58 • select distinct model_name, unit_price from fact_transactions as t
59   left join dim_model as m
60   on t.idmodel = m.idmodel
61   order by unit_price
62   limit 1;
63
64
65 -- Q4--END
66
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```

model_name	unit_price
3210	14.00

ANALYSIS

Model 3210 is the lowest-priced cellphone at 14.00, making it an affordable option for customers.

Q5 What is the average price of models belonging to the top 5 manufacturers by sales quantity?

SQL QUERY

```

select t.idmodel, avg(totalprice) as avgprice
from fact_transactions as t
left join dim_model as m
on t.idmodel = m.idmodel
where idmanufacturer in (select idmanufacturer from
                        (select m.idmanufacturer,
                          manufacturer_name, avg(quantity),
                          sum(totalprice) from
                            fact_transactions as t
                            left join dim_model as m
                            on t.idmodel = m.idmodel
                            left join dim_manufacturer as a
                            on m.idmanufacturer= a.idmanufacturer
                          group by idmanufacturer,
                                manufacturer_name
                          order by sum(quantity) desc
                          limit 5) as x)

group by idmodel
order by avgprice desc

```

RESULT

Top 5 manufacturers are calculated on the basis of sales and quantity. and then average price of all the models of those top 5 manufacturers have been displayed.

```

68 -- Find out the average price for each model in the top5 manufacturers in
69 -- terms of sales quantity and order by average price.
70
71 • select t.idmodel, avg(totalprice) as avgprice from fact_transactions as t
72 left join dim_model as m
73 on t.idmodel = m.idmodel
74
75 where idmanufacturer in (select idmanufacturer from
76 (select m.idmanufacturer, manufacturer_name, avg(quantity), sum(totalprice) from fact_transaction
77 left join dim_model as m
78 on t.idmodel = m.idmodel
79 left join dim_manufacturer as a
80 on m.idmanufacturer= a.idmanufacturer
81 group by idmanufacturer, manufacturer_name
82 order by sum(quantity) desc
83 limit 5) as x)
84 group by idmodel
85 order by avgprice desc;
86

```

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Result Grid Filter Rows: Search Export:

idmodel	avgprice
105	833.714286
101	597.952381
130	588.818182
102	535.142857
103	524.714286
104	503.586235
124	478.000000
129	407.181818
123	394.000000
122	343.600000
128	320.000000
111	285.909091
120	219.200000

ANALYSIS

The analysis identifies the top 5 manufacturers based on sales performance and then compares the average prices of their models to understand differences in sales and pricing.

Q6 Which customers spent more than \$500 on average in 2009 and the average amount they spent?

SQL QUERY

```

select customer_name, avg(totalprice) as
avgamt from dim_customer as c
left join fact_transactions as t
on c.idcustomer = t.idcustomer
left join dim_date as d
on d.dte = t.dte
where yr = '2009'
group by customer_name
having avg(totalprice) > 500

```

RESULT

The result shows the customer's names whose average spend is greater than 500 and the average amount they spent in 2009. There are total 6 customers whose average spend is greater than 500.

```

91  -- Q6--BEGIN
92
93
94  -- List the names of the customers and the average amount spent in 2009,
95  -- where the average is higher than 500
96
97  • select customer_name, avg(totalprice) as avgamt from dim_customer as c
98    left join fact_transactions as t
99      on c.idcustomer = t.idcustomer
100  left join dim_date as d
101    on d.dte = t.dte
102  where yr = '2009'
103  group by customer_name
104  having avg(totalprice) > 500;
105
106
107  -- Q6--END
108
109  -- Q7--BEGIN
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```

customer_name	avgamt
Celeste Korando	613.000000
Moore Parlati	623.500000
Shawna Palaspas	569.000000
Laurel Reller	1528.000000
Danica Bruschie	760.000000
Lettie Isenhower	870.000000

ANALYSIS

The analysis shows that six customers spent more than \$500 on average in 2009, making them high-spending customers.

Q7 Which models were among the top 5 by quantity in 2008, 2009, and 2010 simultaneously?

SQL QUERY

```

select idmodel from
    ( select idmodel , sum(quantity) from fact_transactions
      where year(dte) = '2009'
      group by idmodel
      order by sum(quantity) desc
      limit 5) as a
intersect
    ( select idmodel from
      (select idmodel , sum(quantity) from fact_transactions
       where year(dte) = '2008'
       group by idmodel
       order by sum(quantity) desc
       limit 5) as b
intersect
    ( select idmodel from
      ( select idmodel , sum(quantity) from fact_transactions
        where year(dte) = '2010'
        group by idmodel
        order by sum(quantity) desc
        limit 5) as c

```

RESULT

Model -109 is the only model which was highly purchased and was in top 5 simultaneously in 2008, 2009 and 2010.

```

122 -- 7. List if there is any model that was in the top 5 in terms of quantity,
123 -- simultaneously in 2008, 2009 and 2010
124
125 select idmodel from
126 ( select idmodel , sum(quantity) from fact_transactions
127   where year(dte) = '2009'
128   group by idmodel
129   order by sum(quantity) desc
130   limit 5) as a
131 intersect
132 select idmodel from
133 (select idmodel , sum(quantity) from fact_transactions
134   where year(dte) = '2008'
135   group by idmodel
136   order by sum(quantity) desc
137   limit 5) as b
138 intersect
139 select idmodel from
140 ( select idmodel , sum(quantity) from fact_transactions
141   where year(dte) = '2010'
142   group by idmodel
143   order by sum(quantity) desc
144   limit 5) as c

```

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Result Grid

idmodel
109

ANALYSIS

The Model 109 was consistent among the top 5 most purchased models across 2008, 2009, and 2010, showing its strong and consistent customer demand over the three years.

Q8 Which manufacturer had the second-highest sales in 2009 and 2010?

SQL QUERY

```

select * from
(select manufacturer_name, year(t.dte) as yr,
sum(totalprice) as sale
from fact_transactions as t
left join dim_model as m
on t.idmodel = m.idmodel
left join dim_manufacturer as a
on m.idmanufacturer= a.idmanufacturer
left join dim_date as d
on d.dte = t.dte
where yr = '2009'
group by manufacturer_name, year(t.dte)
having sale <> (select top_sale from
                (select sum(totalprice) as top_sale
                 from fact_transactions as t
                 left join dim_model as m
                 on t.idmodel = m.idmodel
                 left join dim_manufacturer as a
                 on m.idmanufacturer= a.idmanufacturer
                 left join dim_date as d
                 on d.dte = t.dte
                 where yr = '2009'
                 group by manufacturer_name
                 order by sum(totalprice) desc
                 limit 1) as x)

```

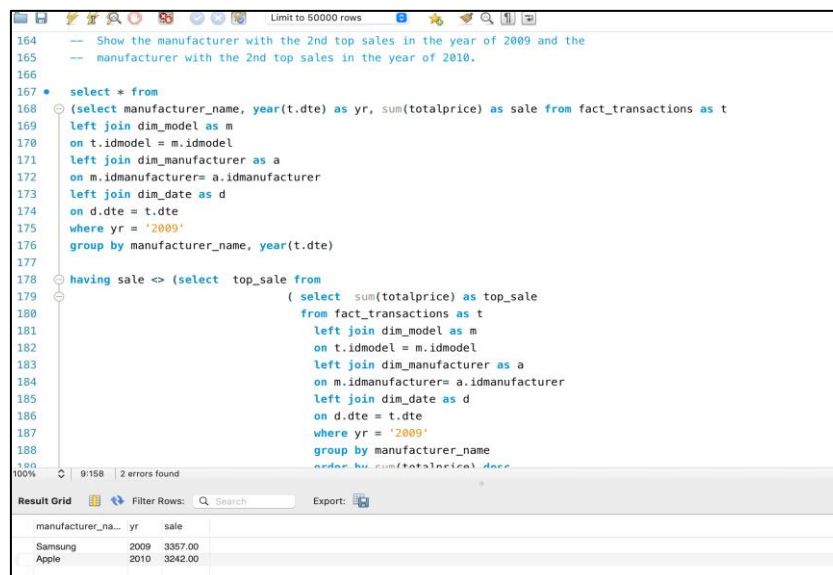
```

order by sum(totalprice) desc
limit 1) as a
union
select * from
(select manufacturer_name, year(t.dte) as yr,
sum(totalprice) as sale
from fact_transactions as t
left join dim_model as m
on t.idmodel = m.idmodel
left join dim_manufacturer as a
on m.idmanufacturer= a.idmanufacturer
left join dim_date as d
on d.dte = t.dte
where yr = '2010'
group by manufacturer_name, year(t.dte)
having sale <> (select top_sale from
                (select sum(totalprice) as top_sale
                 from fact_transactions as t
                 left join dim_model as m
                 on t.idmodel = m.idmodel
                 left join dim_manufacturer as a
                 on m.idmanufacturer= a.idmanufacturer
                 left join dim_date as d
                 on d.dte = t.dte
                 where yr = '2010'
                 group by manufacturer_name
                 order by sum(totalprice) desc
                 limit 1) as x)
order by sum(totalprice) desc
limit 1) as b

```

RESULT

Samsung is the manufacturer with top 2 sales in 2009 and Apple has 2nd highest sales in the year of 2010.



The screenshot shows a SQL query editor with a query window and a results grid. The query is a complex SQL statement using subqueries and joins to find the top 2 sales for each manufacturer in 2009 and 2010. The results grid shows the output of the query, which is a table with columns: manufacturer_name, yr, and sale. The results are as follows:

manufacturer_name	yr	sale
Samsung	2009	3357.00
Apple	2010	3242.00

ANALYSIS

Samsung had the second-highest sales in 2009, while Apple ranked second in 2010. This indicates shift in manufacturer performance.

Q9 Which manufacturers sold cellphones in 2010 but not in 2009?

SQL QUERY

```
select manufacturer_name from
    (select distinct manufacturer_name, yr
     from fact_transactions as t
     left join dim_model as m
     on t.idmodel = m.idmodel
     left join dim_manufacturer as a
     on m.idmanufacturer= a.idmanufacturer
     left join dim_date as d
     on d.dte = t.dte
     where yr = '2010') as x
except
select manufacturer_name from
    (select distinct manufacturer_name, yr
     from fact_transactions as t
     left join dim_model as m
     on t.idmodel = m.idmodel
     left join dim_manufacturer as a
     on m.idmanufacturer= a.idmanufacturer
     left join dim_date as d
     on d.dte = t.dte
     where yr = '2009') as y
```

RESULT

HTC is the manufacturer who sold cellphones in 2010 but not in 2009.

```

240
241
242 -- Show the manufacturers that sold cellphones in 2010 but did not in 2009.
243
244 select manufacturer_name from
245
246 (SELECT distinct manufacturer_name, yr
247 FROM FACT_TRANSACTIONS AS T
248 LEFT JOIN DIM_MODEL AS M
249 ON T.IDMODEL = M.IDMODEL
250 LEFT JOIN DIM_MANUFACTURER AS A
251 on M.IDMANUFACTURER= A.IDMANUFACTURER
252 LEFT JOIN DIM_DATE AS D
253 on d.dte = t.dte
254 where yr = '2010') as x
255 except
256
257 select manufacturer_name from
258 (SELECT distinct manufacturer_name, yr
259 FROM FACT_TRANSACTIONS AS T
260 LEFT JOIN DIM_MODEL AS M
261 ON T.IDMODEL = M.IDMODEL
262 LEFT JOIN DIM_MANUFACTURER AS A
263 on M.IDMANUFACTURER= A.IDMANUFACTURER
264 LEFT JOIN DIM_DATE AS D
265 on d.dte = t.dte
266 where yr = '2009') as y

```

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manufacturer_na...
HTC

ANALYSIS

HTC was the only manufacturer with cellphone sales in 2010 but no recorded sales in 2009. This shows that HTC sale emerged in the during 2010.

Q10 Who are the top 100 customers based on average spending, average quantity purchased and how did their spending change year over year?

SQL QUERY

```

select *, ((avgspend-prev_spend)/prev_spend)
as per_chnge from (select *, lag(avgspend, 1) over (partition by
idcustomer order by yr) as prev_spend from
(select t.idcustomer, customer_name, year(dte) as yr, avg(totalprice) as
avgspend, avg(quantity) as avgqty from dim_customer as c
left join fact_transactions as t
on t.idcustomer = c.idcustomer
where (t.idcustomer,
customer_name) in (select x.idcustomer, customer_name from
(select t.idcustomer, customer_name,
avg (totalprice) as spend
from dim_customer as c
left join fact_transactions as t
on t.idcustomer = c.idcustomer
group by c.idcustomer, customer_name
order by spend desc
limit 100) as x)
group by idcustomer, customer_name, year(dte)) as y) as z

```

RESULT

The result shows top 100 customers their yearly average spend, average quantity purchased and percentage of change in spend YOY.

```

258
259 -- Find top 100 customers and their average spend, average quantity by each
260 -- year. Also find the percentage of change in their spend.
261
262 • select *, ((avgspend-prev_spend)/prev_spend) as per_change
263 from(select *, lag(avgspend, 1) over (partition by idcustomer order by yr) as prev_spend
264 from(select t.idcustomer, customer_name, year(dte) as yr, avg(totalprice) as avgspend, avg(quantity) as avgqty
265 from dim_customer as c
266 left join fact_transactions as t
267 on t.idcustomer = c.idcustomer
268 where (t.idcustomer, customer_name) in (select x.idcustomer, customer_name
269 from
270 (select t.idcustomer, customer_name, avg(totalprice) as spend
271 from dim_customer as c
272 left join fact_transactions as t
273 on t.idcustomer = c.idcustomer
274 group by c.idcustomer, customer_name
275 order by spend desc
276 limit 100) as x)
277 group by idcustomer, customer_name, year(dte) as y) as z;
278
279

```

100% 14:257 2 errors found

Result Grid Filter Rows: Search Export:

idcustomer	customer_name	yr	avgspend	avgqty	prev_spend	per_change
10001	Kallie Blackwood	2004	419.000000	1.0000	20.000000	7.3800000000
10001	Kallie Blackwood	2005	84.000000	1.0000	419.000000	-0.7995226730
10001	Kallie Blackwood	2006	583.000000	1.5000	84.000000	5.9404761905
10001	Kallie Blackwood	2009	381.000000	1.0000	583.000000	-0.3464837050
10002	Johnetta Abdallah	2003	35.000000	1.0000		
10002	Johnetta Abdallah	2004	503.000000	1.0000	35.000000	13.3714285714
10002	Johnetta Abdallah	2006	179.000000	1.0000	503.000000	-0.6441351889
10002	Johnetta Abdallah	2007	396.000000	1.0000	179.000000	1.212905028
10002	Johnetta Abdallah	2008	180.000000	1.0000	396.000000	-0.5454545455

ANALYSIS

The analysis shows the yearly average spending and average quantity purchased for the top 100 customers, along with the year-over-year change in their spending. This helps identify changes in customer spending patterns and purchasing behavior over time.